Model Architecture: Qwen3.5-0.8B

Generated by sg_modelload

July 21, 2026

1 Model Overview

Model Name	Qwen3.5-0.8B
Path	/models/memorization/Qwen3.5-0.8B
Total Parameters	752,393,024
Weight Dtype	bfloat16

2 Forward Pass Algorithm

Algorithm 1 Forward pass of Qwen3.5-0.8B

$h \leftarrow \text{embed_tokens}(x) \in \mathbb{R}^{B \times T \times 1024}$
for $layer \leftarrow 0$ **to** 23 **do**
    $h \leftarrow \text{layer}_{layer}(h)$
end for
$h \leftarrow \text{norm}(h) \in \mathbb{R}^{B \times T \times 1024}$
$\text{logits} \leftarrow \text{lm_head}(h) \in \mathbb{R}^{B \times T \times 248320}$

3 Layer-by-Layer Architecture

All weight tensors and their dimensions are listed below.

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.0	Qwen3_5DecoderLayer	–	–	–
model.layers.0.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.0.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.0.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.0.linear_attn.conv1d.weight	Conv1d	$6144 \times 1 \times 4$	bfloat16	24,576
model.layers.0.linear_attn.in_proj_a.weight	Linear	16×1024	bfloat16	16,384
model.layers.0.linear_attn.in_proj_b.weight	Linear	16×1024	bfloat16	16,384
model.layers.0.linear_attn.in_proj_qkv.weight	Linear	6144×1024	bfloat16	6,291,456
model.layers.0.linear_attn.in_proj_z.weight	Linear	2048×1024	bfloat16	2,097,152
model.layers.0.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.0.linear_attn.out_proj.weight	Linear	1024×2048	bfloat16	2,097,152
model.layers.0.mlp	Qwen3_5MLP	–	–	–
model.layers.0.mlp.down_proj.weight	Linear	1024×3584	bfloat16	3,670,016
model.layers.0.mlp.gate_proj.weight	Linear	3584×1024	bfloat16	3,670,016
model.layers.0.mlp.up_proj.weight	Linear	3584×1024	bfloat16	3,670,016
model.layers.0.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.1	Qwen3_5DecoderLayer	–	–	–
model.layers.1.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.1.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.1.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.1.linear_attn.conv1d.weight	Conv1d	$6144 \times 1 \times 4$	bfloat16	24,576
model.layers.1.linear_attn.in_proj_a.weight	Linear	16×1024	bfloat16	16,384
model.layers.1.linear_attn.in_proj_b.weight	Linear	16×1024	bfloat16	16,384

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.1.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.1.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.1.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.1.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.1.mlp	Qwen3_5MLP	–	–	–
model.layers.1.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.1.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.1.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.1.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.2	Qwen3_5DecoderLayer	–	–	–
model.layers.2.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.2.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.2.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.2.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.2.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.2.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.2.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.2.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.2.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.2.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.2.mlp	Qwen3_5MLP	–	–	–
model.layers.2.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.2.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.2.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.2.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.3	Qwen3_5DecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.3.mlp	Qwen3_5MLP	–	–	–
model.layers.3.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.3.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.3.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.3.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.3.self_attn	Qwen3_5Attention	–	–	–
model.layers.3.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.3.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.3.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.3.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.3.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.layers.3.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.4	Qwen3_5DecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.4.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.4.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.4.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.4.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.4.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.4.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.4.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.4.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.4.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.4.mlp	Qwen3_5MLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.4.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.4.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.4.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.5	Qwen3_5DecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.5.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.5.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.5.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.5.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.5.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.5.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.5.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.5.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.5.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.5.mlp	Qwen3_5MLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.5.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.5.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.5.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.6	Qwen3_5DecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.6.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.6.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.6.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.6.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.6.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.6.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.6.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.6.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.6.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.6.mlp	Qwen3_5MLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.6.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.6.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.6.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.7	Qwen3_5DecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.7.mlp	Qwen3_5MLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.7.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.7.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.7.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.7.self_attn	Qwen3_5Attention	–	–	–
model.layers.7.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.7.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.7.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.7.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.7.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.layers.7.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.8	Qwen3_5DecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.8.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.8.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.8.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.8.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.8.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.8.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.8.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.8.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.8.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.8.mlp	Qwen3_5MLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.8.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.8.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.8.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.9	Qwen3_5DecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.9.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.9.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.9.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.9.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.9.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.9.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.9.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.9.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.9.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.9.mlp	Qwen3_5MLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.9.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.9.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.9.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.10	Qwen3_5DecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.10.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.10.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.10.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.10.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.10.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.10.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.10.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.10.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.10.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.10.mlp	Qwen3_5MLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.10.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.10.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.10.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.11	Qwen3_5DecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.11.mlp	Qwen3_5MLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.11.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.11.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.11.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.11.self_attn	Qwen3_5Attention	–	–	–
model.layers.11.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.11.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.11.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.11.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.11.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.layers.11.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.12	Qwen3_5DecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.12.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.12.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.12.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.12.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.12.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.12.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.12.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.12.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.12.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.12.mlp	Qwen3_5MLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.12.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.12.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.12.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.13	Qwen3_5DecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.13.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.13.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.13.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.13.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.13.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.13.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.13.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.13.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.13.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.13.mlp	Qwen3_5MLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.13.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.13.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.13.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.14	Qwen3_5DecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.14.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.14.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.14.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.14.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.14.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.14.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.14.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.14.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.14.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.14.mlp	Qwen3_5MLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.14.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.14.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.14.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.15	Qwen3_5DecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.15.mlp	Qwen3_5MLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.15.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.15.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.15.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.15.self_attn	Qwen3_5Attention	–	–	–
model.layers.15.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.15.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.15.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.15.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.15.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.layers.15.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.16	Qwen3_5DecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.16.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.16.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.16.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.16.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.16.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.16.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.16.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.16.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.16.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.16.mlp	Qwen3_5MLP	–	–	–
model.layers.16.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.16.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.16.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.16.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.17	Qwen3_5DecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.17.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.17.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.17.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.17.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.17.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.17.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.17.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.17.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.17.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.17.mlp	Qwen3_5MLP	–	–	–
model.layers.17.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.17.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.17.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.17.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.18	Qwen3_5DecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.18.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.18.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.18.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.18.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.18.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.18.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.18.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.18.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.18.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.18.mlp	Qwen3_5MLP	–	–	–
model.layers.18.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.18.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.18.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.18.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.19	Qwen3_5DecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.19.mlp	Qwen3_5MLP	–	–	–
model.layers.19.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.19.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.19.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.19.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.19.self_attn	Qwen3_5Attention	–	–	–
model.layers.19.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.19.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.19.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.19.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.19.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.layers.19.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.20	Qwen3_5DecoderLayer	–	–	–
model.layers.20.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.20.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.20.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.20.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.20.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.20.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.20.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.20.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.20.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.20.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.20.mlp	Qwen3_5MLP	–	–	–
model.layers.20.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.20.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.20.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.20.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.21	Qwen3_5DecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.21.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.21.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.21.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.21.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.21.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.21.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.21.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.21.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.21.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.21.mlp	Qwen3_5MLP	–	–	–
model.layers.21.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.21.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.21.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.21.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.22	Qwen3_5DecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.22.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.22.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.layers.22.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.layers.22.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.22.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.layers.22.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.layers.22.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.22.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.layers.22.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.22.mlp	Qwen3_5MLP	–	–	–
model.layers.22.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.22.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.22.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.22.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.23	Qwen3_5DecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.23.mlp	Qwen3_5MLP	–	–	–
model.layers.23.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.layers.23.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.23.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.layers.23.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.layers.23.self_attn	Qwen3_5Attention	–	–	–
model.layers.23.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.23.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.layers.23.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.23.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.layers.23.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.23.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
lm_head.weight	Linear	248320 × 1024	bfloat16	254,279,680
model	Qwen3_5TextModel	–	–	–
model.embed_tokens.weight	Embedding	248320 × 1024	bfloat16	254,279,680
model.layers	ModuleList	–	–	–
model.norm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024

4 Parameter Count Estimation

Component	Parameters	% of Total
MLP	264,241,152	35.1%
Embedding	254,279,680	33.8%
LM Head	254,279,680	33.8%
Other	189,776,448	25.2%
Attention	44,043,264	5.9%
LayerNorm	52,480	0.0%
Total	**752,393,024**	**100.0%**

4.1 Per-Layer Parameter Count

Layer	Params	Running Total
model.layers.0.input_layernorm	1,024	1,024
model.layers.0.linear_attn	32	1,056
model.layers.0.linear_attn.conv1d	24,576	25,632
model.layers.0.linear_attn.in_proj_a	16,384	42,016
model.layers.0.linear_attn.in_proj_b	16,384	58,400
model.layers.0.linear_attn.in_proj_qkv	6,291,456	6,349,856
model.layers.0.linear_attn.in_proj_z	2,097,152	8,447,008
model.layers.0.linear_attn.norm	128	8,447,136
model.layers.0.linear_attn.out_proj	2,097,152	10,544,288
model.layers.0.mlp.down_proj	3,670,016	14,214,304
model.layers.0.mlp.gate_proj	3,670,016	17,884,320
model.layers.0.mlp.up_proj	3,670,016	21,554,336
model.layers.0.post_attention_layernorm	1,024	21,555,360
model.layers.1.input_layernorm	1,024	21,556,384
model.layers.1.linear_attn	32	21,556,416
model.layers.1.linear_attn.conv1d	24,576	21,580,992
model.layers.1.linear_attn.in_proj_a	16,384	21,597,376
model.layers.1.linear_attn.in_proj_b	16,384	21,613,760
model.layers.1.linear_attn.in_proj_qkv	6,291,456	27,905,216
model.layers.1.linear_attn.in_proj_z	2,097,152	30,002,368
model.layers.1.linear_attn.norm	128	30,002,496
model.layers.1.linear_attn.out_proj	2,097,152	32,099,648
model.layers.1.mlp.down_proj	3,670,016	35,769,664
model.layers.1.mlp.gate_proj	3,670,016	39,439,680
model.layers.1.mlp.up_proj	3,670,016	43,109,696
model.layers.1.post_attention_layernorm	1,024	43,110,720
model.layers.2.input_layernorm	1,024	43,111,744
model.layers.2.linear_attn	32	43,111,776
model.layers.2.linear_attn.conv1d	24,576	43,136,352
model.layers.2.linear_attn.in_proj_a	16,384	43,152,736
model.layers.2.linear_attn.in_proj_b	16,384	43,169,120
model.layers.2.linear_attn.in_proj_qkv	6,291,456	49,460,576
model.layers.2.linear_attn.in_proj_z	2,097,152	51,557,728
model.layers.2.linear_attn.norm	128	51,557,856
model.layers.2.linear_attn.out_proj	2,097,152	53,655,008
model.layers.2.mlp.down_proj	3,670,016	57,325,024
model.layers.2.mlp.gate_proj	3,670,016	60,995,040
model.layers.2.mlp.up_proj	3,670,016	64,665,056
model.layers.2.post_attention_layernorm	1,024	64,666,080
model.layers.3.input_layernorm	1,024	64,667,104
model.layers.3.mlp.down_proj	3,670,016	68,337,120
model.layers.3.mlp.gate_proj	3,670,016	72,007,136
model.layers.3.mlp.up_proj	3,670,016	75,677,152
model.layers.3.post_attention_layernorm	1,024	75,678,176
model.layers.3.self_attn.k_norm	256	75,678,432
model.layers.3.self_attn.k_proj	524,288	76,202,720
model.layers.3.self_attn.o_proj	2,097,152	78,299,872
model.layers.3.self_attn.q_norm	256	78,300,128
model.layers.3.self_attn.q_proj	4,194,304	82,494,432
model.layers.3.self_attn.v_proj	524,288	83,018,720
model.layers.4.input_layernorm	1,024	83,019,744
model.layers.4.linear_attn	32	83,019,776
model.layers.4.linear_attn.conv1d	24,576	83,044,352
model.layers.4.linear_attn.in_proj_a	16,384	83,060,736
model.layers.4.linear_attn.in_proj_b	16,384	83,077,120
model.layers.4.linear_attn.in_proj_qkv	6,291,456	89,368,576
model.layers.4.linear_attn.in_proj_z	2,097,152	91,465,728
model.layers.4.linear_attn.norm	128	91,465,856
model.layers.4.linear_attn.out_proj	2,097,152	93,563,008
model.layers.4.mlp.down_proj	3,670,016	97,233,024
model.layers.4.mlp.gate_proj	3,670,016	100,903,040
model.layers.4.mlp.up_proj	3,670,016	104,573,056